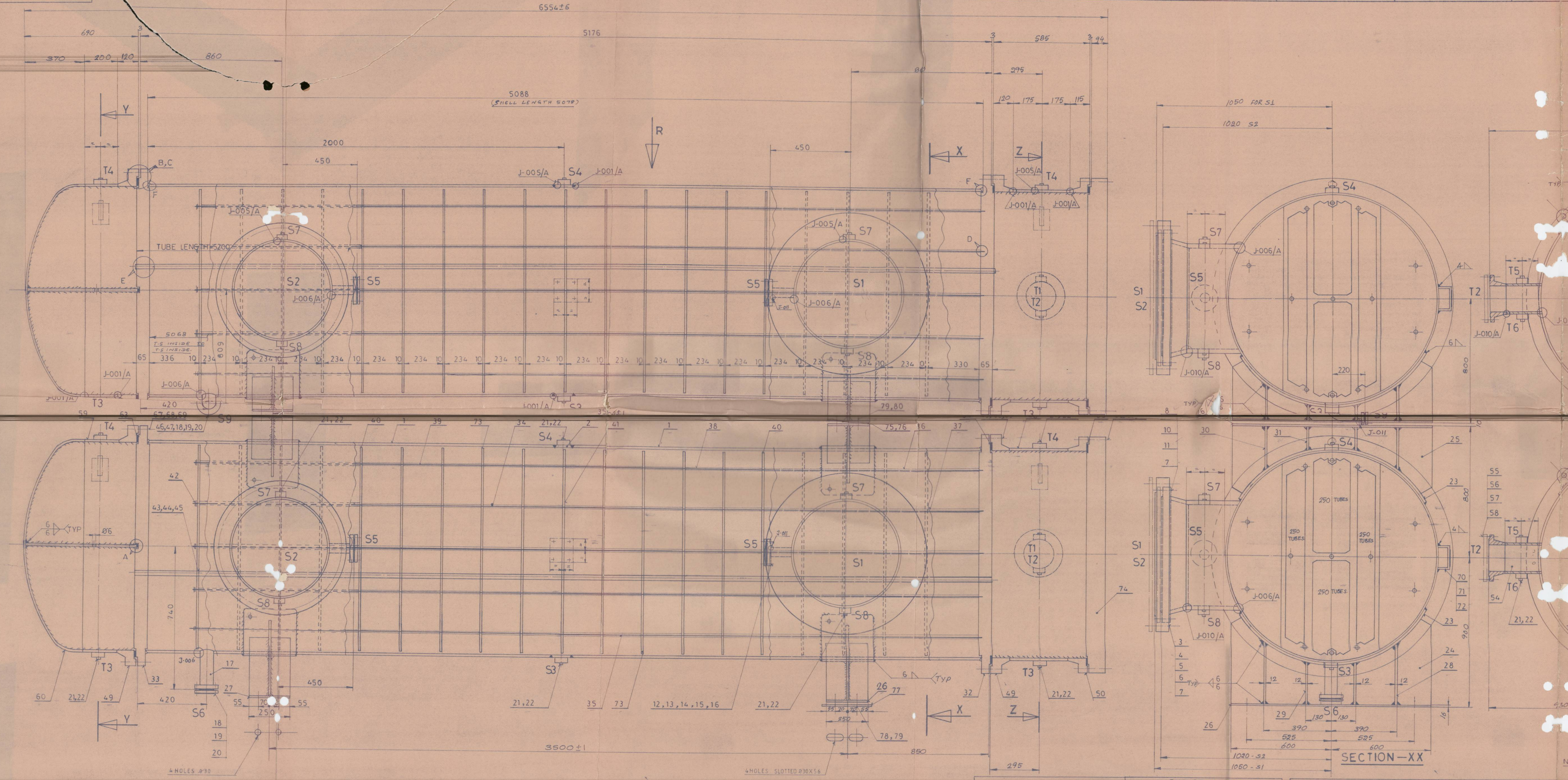
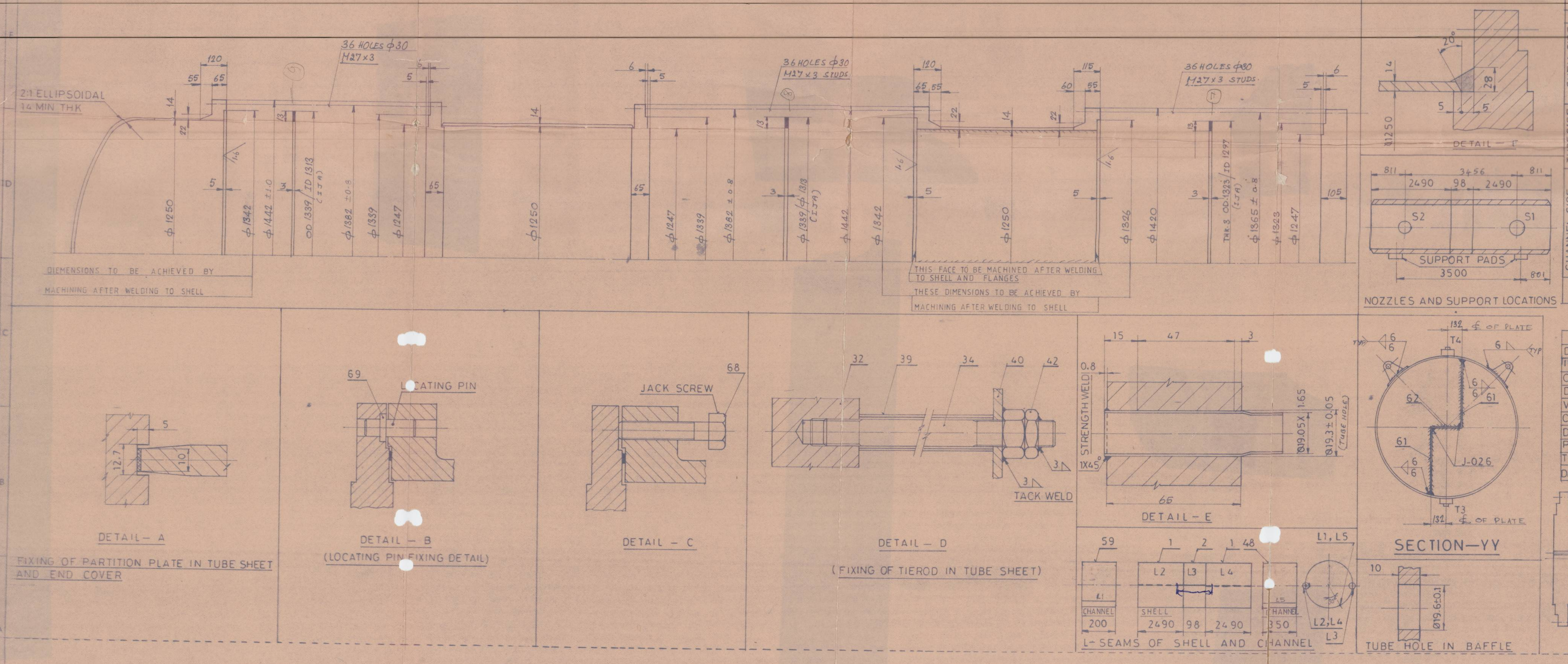
(ALL DIMENSIONS ARE IN mm)





NOTES:-

1. THE PROJECTION INSIDE THE SHELL BE GROUND SMOOTH.
2. FLANGE HOLES SHALL STRADDLE THE VESSEL CENTRE LINES.
3. FOR PAINTING SEE THE SPECIFICATION HE11008.
4. TOLERANCES ON ALL DIMENSIONS UNLESS SPECIFIED AS PER TEMA 'R'
5. HYDRAULIC TEST AS PER HY0852061.
6. TUBE WALL THINNING SHALL BE 3 TO 5%.
7. M6 TAPPED HOLES ARE TO BE PROVIDED ON RF PADS
8. APPROVED QUALITY PLAN NO. HYD/SEC(AUX)06/498 DT10/1/91; REV01 (ONE SHEET)
9. PART LIST NO 4-16680-01810 AND DOC. NO. HE200021.
10. WELD DETAILS AND WPS NOS AS PER DRG.NO.2-16633-00010
11. CUSTOMER DRG NO 3-16680-00362
12. TUBE LAYOUT DRG NO 2-16680-00195

BILL OF MATERIALS

	DESCRIPTION	MATERIALS
SHELL SIDE	SHELL	SA 516/70
	GIRTH FLANGES	SA 266 CL2
	NOZZLE PIPES	SA 106/B
	NOZZLE FLANGES	SA 105
	SADDLES	IS 2062
	GASKETS	IJA
TUBE SIDE	TUBE	SA 249 TP304
	TUBE SHEETS	F 304
	BAFFLES	IS 2062
	TIE RODS	IS 1978 GR.St
CHANNEL SIDE	SHELL	SA 516/70
	DISHED END	SA 516/70
	GIRTH FLANGES	SA 266 CL2
	NOZZLE PIPES	SA 106/B
	NOZZLE FLANGES	SA 105
	GASKETS	IJA
	STUDS & NUTS	SA193-B7/194-2H

DESIGN DATA

DESCRIPTION	SHELL SIDE	TUBE SIDE
DESIGN PRESSURE Kg/Cm²(g)	4	6
TEST PRESSURE Kg/Cm²(g)	6	9
DESIGN TEMPERATURE °C	200	60
CORROSION ALLOWANCE mm	3.2	3.2
STRESS RELIEVING	NO	NO
RADIO GRAPHY	SPOT	SPOT
NO OF PASSES	1	4
MEDIUM	AIR	WATER
JOINT EFFICIENCY	0.85	0.85
INSPECTION	M/S PDIL	
SURFACE AREA PER COOLER	293	
NO OF TUBES PER COOLER	1000	
TUBE LENGTH mm (EFF)	5068	
TUBE SIZE ODXTHK	19.05X1.65	
PITCH / TYPE mm	1" Δ	
NO OF BAFFLES	3	
BAFFLE PITCH mm	1220	
EMPTY WEIGHT Kg (BOTH COOLERS)	22300.0	
WORKING WEIGHT Kg "	27500.0	
FLOODED WEIGHT Kg "	38600.0	
NO OF DUMMY BAFFLE / SPACING	4	

SHELL SIDE NOZZLES

DESCRIPTION	MARK	SIZE	SERIES
INLET (AIR)	S1	26"	ANSI 150# WNRF
OUTLET (AIR)	S2	24"	ANSI 150# WNRF
DRAIN	S3	1"	ANSI 6000#NPT COUPLING
VENT	S4	1"	ANSI 6000#NPT COUPLING
CHEMICAL CLEANING	S5	1 1/2"	ANSI 300# SORF
DRAIN TO SEPARATOR	S6	2"	ANSI 150# LWNRF
PR. GAUGE	S7	1"	ANSI 6000#NPT COUPLING
THERMO WELL	S8	1"	ANSI 6000#NPT COUPLING
DRAIN TO SEPARATOR	S9	2"	ANSI 150# SORF

TUBE SIDE NOZZLES

DESCRIPTION	MARK	SIZE	SERIES
WATER INLET	T1	6"	ANSI 150# WNRF
WATER OUTLET	T2	6"	ANSI 150# WNRF
DRAIN	T3	1"	ANSI 6000#NPT COUPLING
VENT	T4	1"	ANSI 6000#NPT COUPLING
PR. GAUGE	T5	1"	ANSI 6000#NPT COUPLING
THERMO WELL	T6	1"	ANSI 6000#NPT COUPLING

HE DESIGN COPY

ARRANGEMENT :- TWO COOLERS IN PARALLEL

DESIGN CODE:- ASME SEC-VIII-DIV-I 1989 ADD A'89: TEMA 'R' 1988.

W.O.NO. 10466 41866 - CA114 VESSEL NO. 0367/A - TOP, 0367/B - BOTTOM

TYPE OF PRODUCT OR NAME OF CUSTOMER/PROJECT: TCL BABRALA AIR

BHARAT HEAVY ELECTRICALS LTD
HYDERABAD

DRN. R.S.R. CHD. M.B.D. APPD. V.C.K.

WEIGHT (Kg) 22300.0

TITLE 1ST STAGE COOLE…

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